Storing data in the cloud	Storing data locally
You do not have to pay for the hardware to store the data.	You will need to purchase the hardware to store your data. These costs can be high if you have a lot of data to store.
You do not have the responsibility for the security of the hardware that stores the data. However, you are relying on a third party making sure that your data is safe. If they do not do this correctly, they may put your data at risk.	You will need to make sure that your data storage is well maintained and that it is regularly checked and updated for security issues.
You can access the data and resources using different devices from anywhere that you have an internet connection. However, if your connection fails or is not available, you will not be able to access your data. If your connection is slow, this can also mean that accessing your data is slow.	You do not need an internet connection to access your data, therefore it is available at all times that you have local access to that data.
You can increase and decrease the amount of storage space you need very easily, when you need to.	You might have redundant hardware, that you have paid for, that isn't used as the amount of data that you need to store has decreased.

Table 3.3: Comparisons of storing data in the cloud and locally

ACTIVITY 3.10

Do you store any data in cloud storage? If so, do you ever worry about doing this? Discuss with a partner about the advantages and disadvantages of storing data in the cloud, compared to local storage. Reach a conclusion about whether you think the risk is worth it.

As part of your discussion research at least one more advantage and one more disadvantage and make note of these. You can then share your thoughts from your discussion with the rest of your class and see who thinks cloud or local storage is better, and why.

3.10 Network hardware

At the beginning of this chapter, you learnt about the hardware required within a computer and local to a computer that enable the temporary and permanent storage of data. You have now learnt that data can be stored remotely, in a cloud storage system, but that an internet connection is required to access the data stored in the cloud. The hardware required for this connection creates a **network**. You will now learn what hardware is required to create a connection to other devices and the internet.

When two or more devices are connected, this is called a network. Some basic hardware devices are required to create a network, these are a **network interface card (NIC)**, a **router** and you may need some **transmission media**.

KEY WORDS

network: computers and devices that are joined together using cables or wireless technology.

network interface card (NIC): the component in a computer that is used to connect it to a network.

router: a network component that examines a data packet to obtain its destination address and then forwards the packet to this address.

transmission media: the cables that are used to transfer data in a network.

Within your computer or device is a piece of hardware called a NIC. This may be a wired or wireless NIC. This means that either a cable will be plugged into the hardware to connect to a network, or it will have a wireless component that allows it to make a wireless connection. Most modern NICs are wireless NICs (WNICs).

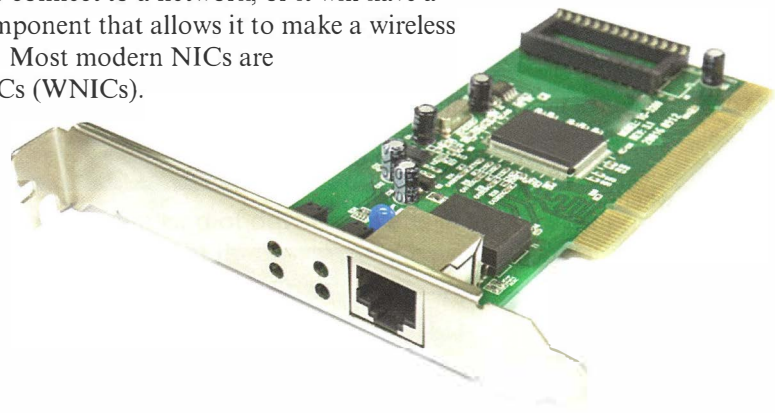


Figure 3.15: A network interface card (NIC)

Each NIC is given an address called a **media access control (MAC) address**. This address can be used to identify a device on a network.

A MAC address is assigned by the manufacturer and is never normally changed. It will remain assigned to the device whilst it is in use. A MAC address is made up of two main parts. These are the manufacturers ID code and the Serial ID code for the device. This creates a unique address for each device. The address is represented as hexadecimal and normally has six pairs of numbers, separated with either a dash or a colon, for example:

34:4D:EA:89:75:B2

When a device with a network interface card transmits data across a network, it will send the MAC address of that device with the data. This means that it is possible to tell, from inspecting the data, which device on the network the data was sent from.

KEY WORD

media access control (MAC) address: the unique address that is given to a NIC by the manufacturer.

ACTIVITY 3.11

See if you can find out the MAC address of one of your devices, such as a computer, laptop or a mobile phone.

If you do not know how to find it, you can use the internet to find a tutorial.

When data is sent across a network, a hardware device can be used to make sure that the data arrives at the correct destination. You can use a hardware device called a router to do this. Will a router have a NIC? Does this mean it will have a MAC address? See if you can find out! Each device that wants to connect to the internet will also need to do this by connecting to the router. This could be a wired or wireless connection.

A router is a hardware device that reads each **data packet** that it receives. Details about data packets and their structure was covered in Chapter 2. The router reads the data packets to find out the destination of the data. It can use the MAC address to do this,

KEY WORD

data packet: a unit of data that is used to transmit data across a network.

but that will normally only be used if the data is sent to another device that is part of the same local network (this is a network that is contained within a specific area, such as your house). As a router is also used to connect a device or a local network to the internet, it is the internet protocol (IP) address of the destination device that is normally read and used to forward the data to its correct destination.

This IP address can also be used to locate your device on a network, and it is also a unique address. It is different in structure to a MAC address, as it is assigned by the network, rather than by the manufacturer. In a local network, such as a home network, it is often the router that assigns the IP address to a device, when it is connected to the router.



Figure 3.16: A home router

An IP address can have different structures, such an IPv4 or IPv6 structure. IPv4 was the first structure created for an IP address. This structure was created back in 1983 and uses 32-bits to create each numerical IP address. Even though it is now an old structure, there are still many devices that have an IPv4 address. An example of an IPv4 address is:

12.244.233.165

As more and more devices started connecting to networks, the number of available IPv4 addresses started to rapidly decline. As each IP address needs to be unique to the device, they were rapidly running out, due to the number of devices worldwide. IPv6 was created as result, to create more available IP addresses. IPv6 uses 128-bits to create an IP address. IPv6 addresses can also contain hexadecimal characters. This created much more possibilities for IP addresses and allowed them to remain unique to the device. An example of an IPv6 address is:

2001:0db8:0000:0000:0000:ff00:0042:7879

You will also notice that an IPv4 address is separated by full stops, but an IPv6 address is separated by colons.

An IP address can also be either **static** or **dynamic.** Most devices are assigned a dynamic IP address. This means that the address can change each time the device is connected to a network. The device may be given that IP address for a single session

KEY WORDS

static IP: an IP address that does not change each time the device connects to a network.

dynamic IP: an IP address that can change each time the device connects to a network.

that it connects to the router, or it may be given it for several sessions before it changes, but it will change at some point. It is possible to obtain a static IP address for your device. This means that the IP address for your device will be the same every time that you connect your device to a network, it will not change, until you request it to be changed, it will always remain the same. It is normally your Internet Service Provider that will provide you with a static IP address for your device.

ACTIVITY 3.12

Can you find out what kind of devices a static IP address is normally used for? Can you also find out what the benefit is of having a static IP address?

SUMMARY

Hardware is all the physical components that you can see and touch in a computer system.
The CPU is the component in a computer system that processes all the instructions and data. It uses an instruction set to do this.
The CPU has several component parts, these include the ALU, CU, PC, MAR, MDR, CIR, ACC and buses.
The CPU is responsible for carrying out the fetch-decode-execute cycle.
The number of cores, the clock speed and the cache size can all affect the performance of the CPU.
Computers also need input and output devices to be able to provide a system with data, then obtain the results when it has been processed.
Sensors are a type of input device that can be used to capture data from the immediate surrounding environment.
A computer has both primary and secondary storage.
Primary storage is directly accessed by the CPU and consist of RAM and ROM.
Secondary storage is not directly accessed by the CPU and there are three types, magnetic, optical and solid-state.
Virtual memory is created by portioning part of the hard drive. It is used when RAM is full.
Cloud storage is when data is stored remotely on servers and storage that is owned and maintained by a third party.
A network is created when two or more devices are connected.
A device needs a NIC to connect to a network. Each NIC has a MAC address.
A router is used in a network to forward packets of data to their correct destination. A router can assign an IP address to each device.
There are two types of IP address, IPv4 and IPv6.